

Visualising Public Transport Service Inequality Across Victorian LGAs

FIT5147: Data Exploration and Visualisation

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Introduction

This project focuses on inequality in public transport services across different Local Government Areas (LGAs) in Victoria. The main message of this interactive narrative visualization is that public transport access should not be judged only by whether an area has stations, stops, or routes. Some areas may look well covered by the public transport network on the map. However, people living there may still find it hard to rely on public transport in daily life. This can happen when services are not frequent, operating hours are short, weekend services are limited, or there are not enough choices of transport modes. Therefore, this project does not only look at whether public transport infrastructure exists. It focuses more on whether these services are useful, reliable, and diverse in real life.

The main point of this visualization is that public transport inequality is a multi-dimensional issue. It can change across places, transport modes, and time patterns. This project compares public transport service gaps across different LGAs by using several indicators. These include the coverage gap, frequency gap, service span gap, weekday-weekend gap, mode diversity gap, and overall inequality score. The map is used to show the spatial pattern of service inequality across Victoria. The ranking chart highlights the most underserved areas under the selected indicator. The following linked views then explain why a certain LGA has a high inequality score. Through this structure, the project does not only show which areas have serious service gaps. It also explains where these gaps come from.

The target audience of this project includes the public, local community members, and non-expert policy-related readers who are interested in public transport equity in Victoria. These users may not have a background in GTFS data, transport planning, or spatial analysis. However, they can usually understand the importance of public transport services through daily experience. For example, they may think about whether people can easily travel to work, school, or social activities. For these users, raw timetable data and complex technical indicators are not easy to understand directly. They need a clearer and more visual way to understand service differences between different areas.

Design Process:

The design process of this project follows the Five Design-Sheet methodology. The overall goal is not only to show a set of public transport indicators. Instead, the project organizes the public transport service gaps between Victorian LGAs into an explorable narrative visualization. Because the target audience is mainly non-expert users, the design process focuses on three key questions:

1. How can users quickly see where inequality exists across Victoria?
2. How can users understand why a certain area is seen as underserved?
3. How can interaction help users explore the areas, transport modes, and indicators they care about?

Based on Munzner's what-why-how framework (Munzner, 2014):

- the “what” of this project is the public transport service indicators aggregated by LGA. These include the coverage gap, frequency gap, service span gap, weekday-weekend gap, mode diversity gap, and overall inequality score.
- The “why” is to help users identify, compare, filter, and explain public transport service gaps across different areas.
- The “how” is achieved through a combination of multiple linked views. These include a choropleth map, ranking bar chart, LGA profile, service span chart, dumbbell chart, bubble chart, and conclusion matrix.

Sheet 1 (See Appendix 1):

Sheet 1 is mainly used for brainstorming and selecting possible design directions. At the beginning, I grouped the possible visual approaches into several categories:

- The first category is spatial distribution visualization. It is used to show geographic differences between different LGAs.
- The second category is ranking and comparison visualisation. It is used to identify the areas with the most obvious service gaps.
- The third category is time structure visualisation. It is used to show issues such as service operating hours and differences between weekdays and weekends.

- The fourth category is multi-indicator explanation visualisation. It is used to break down the different factors behind the overall inequality score.
- The fifth category is summary visualisation. It is used at the end of the story to summarise the main patterns across different regions and transport modes.

The focus of this stage is not to decide the final interface immediately. Instead, it is to choose the visual forms that can best answer the project questions. Therefore, the design direction changed. It moved away from simply thinking about how to show all the data. Instead, it became more focused on exploring where public transport inequality exists.

Sheet 2:

Sheet 2 is a time-first design titled The Rhythm of Inequality. This design places time structure at the center of the narrative. It focuses on whether public transport services are truly usable during the day and across a week. Users first choose the area they want to view. Then, they can understand whether the service supports daily travel by looking at service frequency in different time periods, weekday and weekend differences, and service span. Because time data has a clear order (Aigner et al., 2023), line charts, timelines, and heat maps can keep the time structure of service changes. They are more suitable than a single summary indicator for showing issues such as limited evening services or reduced weekend services.

However, the problem with this design is that the spatial pattern is placed in a less important position. This project focuses on the unequal distribution of public transport services between Victorian LGAs. If users can only see time changes at the beginning, it may be difficult for them to build an overall understanding of the whole state. Therefore, the time-based perspective from Sheet 2 was kept in the final design, but only in the service span chart and the weekday-weekend comparison. It was not used as the main entry point of the whole page.

Sheet 3:

Sheet 3 is a geography-first design titled The Geography of Inequality. This design uses the map as the main entry point. It highlights the spatial distribution of public transport service gaps. It also uses a bubble chart to show the relationship between coverage and intensity. A map is suitable for LGA-

level data because LGAs naturally have geographic locations. Users can directly connect service gaps with places they know. Based on visual variable theory, position is the most natural visual form for spatial data. Color lightness or color intensity is also suitable for showing an ordered gap score (Munzner, 2014). The strength of this design is that the spatial story is clear. Users can first see which areas are darker and have stronger gaps. Then, they can further explore the reasons behind these gaps. However, the map also has limitations. Large regional LGAs may look more important because they take up more space on the map. This may lead to attention bias.

In the final design, I kept the map as the entry point of the narrative. However, I added a ranking chart and linked detail views to reduce the limitations of the map. This allows LGAs with small land areas but clear service gaps to be identified more clearly.

Sheet 4:

Sheet 4 is a gap-first design titled *The Structure of Inequality*. This design is more focused on explanation and analysis. It uses the ranking of underserved LGAs, the overall inequality score, and the score breakdown as the main parts of the interface. The strength of this design is that it has a clearer problem focus. It first uses the ranking to help readers identify which LGAs have the most obvious public transport service gaps. In other words, it shows which areas need more attention. Then, the score breakdown further explains the overall inequality score. It helps readers see that a high score is not a black-box result. Instead, it is caused by several factors, including coverage, frequency, service span, weekday/weekend gap, and mode diversity. Therefore, this design not only shows which areas have more serious problems. It also explains where these problems mainly come from.

However, its weakness is that the page may look too indicator-based if there is no map or narrative text. This may make the design feel less clear for general users. The final design did not use Sheet 4 alone. Instead, it combined the ranking and profile explanation from Sheet 4 with the spatial entry point from Sheet 3.

Sheet 5:

Sheet 5 is the final implemented design, titled *Public Transport Inequality Explorer*. It is not a completely new design. Instead, it is the final realization created by combining and simplifying ideas

from Sheet 2, Sheet 3, and Sheet 4. The final structure follows a narrative path from overview to detail, and then back to summary:

Users first use the choropleth map to see the distribution of service gaps across the whole state. Next, users use the ranking bar chart to identify the most underserved LGAs under the current filter settings. Then, users can select up to three LGAs by clicking the map or the ranking chart. After that, they can compare the specific gaps between these areas through the LGA profile, service span chart, weekday-weekend comparison, and coverage-intensity bubble chart. Finally, the conclusion matrix summarizes the local observations again. It presents the main findings by region and transport mode.

This structure follows Shneiderman's idea of "overview first, zoom and filter, then details-on-demand" (Shneiderman, 1996). It also fits the needs of the target audience of this project: It first helps non-expert readers build an overall impression. Then, it gradually provides more explanation and comparison.

visualization type:

For the final choice of visualisation types, I did not place all information into one complex chart. Instead, I used multiple linked views to support different tasks. The reason for this is that public transport inequality has many layers. A single chart would find it difficult to show all of these clearly at the same time. Therefore, in the final design, each view answers one relatively clear question. These views are then connected through interaction. This reduces the information load of each chart. It also helps readers gradually understand the different sources of public transport inequality.

Visual Variables:

For the choice of visual variables, I tried to use visual forms that are easy to understand and compare. The map uses color lightness to show the gap score. A darker colour means a larger service gap. This helps readers quickly see which areas have more obvious problems without reading every number carefully.

The ranking chart and LGA profile mainly use bar length to show value size. This is because length is easier to compare than area or angle. The bubble chart uses the x-axis and y-axis to show coverage

and service intensity. The size of each point is used to show service scale as extra information. For the LGAs selected by users, I use consistent highlight colours across all linked views. This helps readers track the same area across different charts and reduces confusion.

Color Palette:

For the choice of color palette, I used a light warm background. The aim is to reduce the strong brightness of a pure white page. It also keeps enough contrast for the map and charts. The map uses a sequential color scale to show the continuous inequality score. The interactive highlight color is used to mark the LGA selected by the user. Color is not used too much for decoration. Instead, it supports data separation, status feedback, and visual hierarchy. Based on Tufte's design principle, visual elements should mainly support data expression, rather than create meaningless visual noise (Tufte, E. R. (2001)). Therefore, I avoided using too many patterns, shadows, or unnecessary decorative elements. This helps users focus on the map, ranking, gap breakdown, and comparison relationships.

Typography & Layout:

The design of typography and layout also focuses on non-expert users. The page uses large titles and clear section labels. This helps users understand which stage of the story they are reading. For the titles, we use short narrative sentences instead of only technical chart names. This can lower the entry barrier. It helps users understand the question behind each chart before they look at the data details.

For the layout, the control panel is placed at the top. This allows users to choose the region, mode, metric, and Top N before entering the story. The map and ranking chart are placed in the first main view group because they are used to build the overview and show priority areas. The later views are arranged in a card-based grid. They keep consistent titles, explanations, and chart areas. This layout has a clear visual hierarchy. It also follows the principle of proximity, where elements placed close together are more likely to be seen as related (Johnson, 2014).

Narrative Style:

For the narrative style, this project uses guided exploration. It is not a fully linear slideshow or a fully open dashboard. Users can choose filters and LGAs by themselves. At the same time, the story beats

on the page guide them to understand the issue in this order: spatial pattern, ranking, local explanation, time difference, coverage and intensity, and summary.

This hybrid style is suitable for the target audience of this project. General readers need text guidance to understand what the indicators mean. They also need some freedom to interact with the page and check the areas they care about. The role of the narrative text is not to replace the charts. Instead, it builds an explanation frame for the charts. It helps users understand why these indicators together show public transport usability.

Human Visual System:

From the perspective of the human visual system, the final design tries to avoid showing too much complex information at one time. The page does not put all indicators into one complex chart. Instead, it uses several simple views to support different tasks. The map is used for spatial location. The bar chart is used for ranking and comparison. The profile is used to break down the indicators. The time chart is used to show service periods. The bubble chart is used to show a two-dimensional relationship. The matrix is used to summarise the findings. This layered structure can reduce cognitive load. It helps users build understanding step by step. Visual hierarchy, colour highlights, spatial grouping, and a consistent card structure also help users quickly identify the main information and the supporting explanations.

Compare to DVP Part 1

Compared with the design in DVP Part 1, the final implementation made several changes. First, some early sketches used day type as a global filter. In the final version, the weekday-weekend difference is shown in a separate comparison view instead. This is because too many top controls may make the page harder for general users to understand. Showing this gap as a separate view better supports the narrative order from overview to explanation. Second, the bubble chart was changed from using population size to using GTFS stop events. This makes the chart more directly related to public transport supply. A larger bubble now means more service activity, not just a larger population.

Overall, these changes do not move away from Sheet 5. They refine it based on data availability, interaction consistency, and user understanding.

Implementation

Technical Implementation

The final interactive narrative visualisation of this project is built with HTML, CSS, and vanilla JavaScript. D3.js v7 is used as the main visualisation library. The whole project has three core files: index.html, style.css, and main.js.

The biggest technical challenge of this project was data preparation and indicator calculation in the early stage. The original GTFS schedule data was large. Public transport information was also split across several files, such as stops, routes, trips, stop_times, and calendar. Because of this, the data could not be used directly for visualisation. It had to be cleaned, filtered, and combined first. During data preparation, I checked missing values, NA values, and records that were not suitable for analysis. This helped reduce the effect of abnormal data on later calculations. After that, I used common fields in the GTFS files to do several joins. This connected information about stops, routes, trips, service dates, and stop times. It then formed a data structure that could be aggregated by LGA and transport mode.

After completing the joins, I calculated a set of summary metrics for the visualisation. These metrics include the number of stops, number of routes, number of trips, weekday and weekend stop events, first service hour, last service hour, service span hours, stop density, route density, and events per stop. I also calculated several gap indicators. These include the coverage gap, frequency gap, service gap, weekend-weekday gap, mode diversity gap, and overall inequality score. Because the final website needs to respond quickly to user interactions, the raw GTFS data could not be processed directly in the D3 page. Therefore, I completed the complex data processing before building the webpage. The processed data was exported as a simplified lga_metrics.csv file and an LGA GeoJSON file. This reduced the workload on the front end. It also helped the D3 visualisation support filtering, map clicking, and linked view updates more smoothly.

Compared with the final design in Sheet 5, the actual implementation keeps the same overall narrative structure. It starts with a spatial overview, then moves to ranking comparison. After that, it provides detailed explanation for the selected LGAs. Finally, it returns to a more general summary of patterns.

Interactive Narrative Visualization Implementation

The main reading path of the visualisation starts from the control panel (see Figure 1 below). The control panel provides four main filter dimensions: region type, mode, metric, and Top N. Users can choose to view all areas, Inner Melbourne, Outer Melbourne, or Regional Victoria. They can also switch between Bus, Train, Tram, and All modes. The metric selector allows users to switch between overall inequality score, coverage gap, frequency gap, service gap, and weekend-weekday gap. This design allows users to reorganise the data story from different perspectives. They do not only read one fixed conclusion passively. The text on the page also clearly tells users that the map is the entry point of the story. A darker colour means a stronger service gap under the current metric.

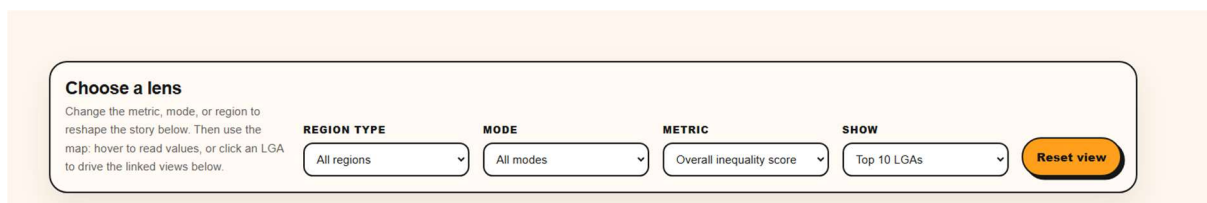


Figure 1 Control Panel

The first group of core views (see Figure 2 below) includes a choropleth map and a ranked bar chart. The map shows the spatial distribution of LGAs in Victoria. It helps users first identify whether public transport service gaps are concentrated in certain areas. Color intensity is used to show the currently selected inequality indicator. This allows users to quickly compare the relative gaps between different LGAs. The map supports a hover tooltip. When users move the mouse over an LGA, they can see information such as the LGA name, region type, current mode, current metric value, number of stops, number of routes, and whether the area has service. The map also supports D3 zoom and pan. Users can zoom in to view dense areas such as Inner Melbourne, or drag the map to check local spatial patterns.

The ranking chart on the right works as a complement to the map. The map answers where the gaps are, while the ranking chart answers which areas are the most serious. The bar chart ranks LGAs by the current metric. It shows the Top 5, Top 10, or Top 20 based on the Top N control. Users can select an area by clicking an LGA on the map. They can also click a bar in the ranking chart. The two views use the same selection colours for linked highlighting. This helps users clearly see the relationship

between a place on the map and its position in the ranking. This design supports users in moving from the broad spatial pattern to specific local issues.

To support comparison, the final implementation allows users to select up to three LGAs. After users select more than one area, the following panels change from a single-area explanation to a comparison view. The three selected LGAs are shown with different colours. These colours stay consistent across the map, ranking chart, profile, service span chart, weekday-weekend chart, bubble chart, and conclusion matrix. This interaction design allows users to compare the sources of inequality in different areas. It does not only focus on the area with the highest score. For example, one LGA may mainly have a coverage problem. Another LGA may mainly have low service frequency or an early last service time.

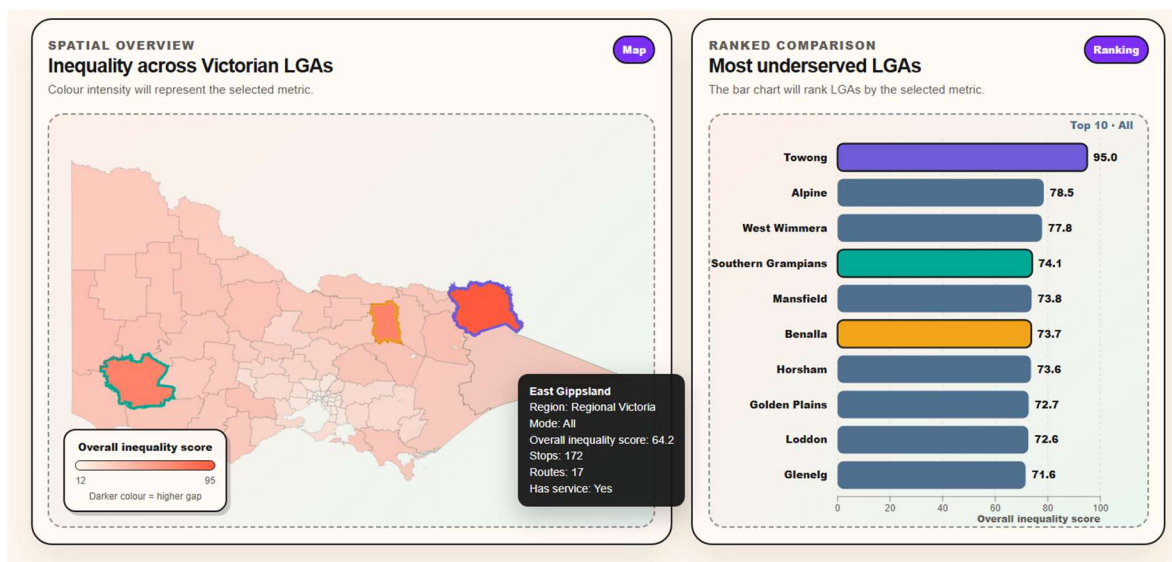


Figure 2 Choropleth Map (Left) & Ranked Bar Chart (Right)

In the second group of views (see Figure 3 below), the LGA profile panel is used to explain the sources of inequality in the selected area. It shows indicators such as coverage gap, frequency gap, service gap, weekend-weekday gap, and mode diversity gap. This helps users understand that a high inequality score is not caused by only one factor. This panel plays an “explanation” role in the narrative. The ranking chart tells users which areas are the most underserved. The profile panel then explains why these areas are underserved. When users select multiple LGAs, this panel changes into a comparison view. This helps users compare the indicator structure of different areas side by side.

The service span chart next to it further turns “inequality” into daily user experience. It shows the first service time and last service time for Bus, Train, and Tram. It also adds a 9 pm reference line as a guide for evening service. The purpose of this view is not only to show whether service exists. It also shows whether the service covers a long enough part of the day. If an LGA only has bus service that ends early, or if one transport mode has no recorded service, users can directly see the limits of daily travel in that area. When users select multiple LGAs, this chart is also divided into several comparison blocks. This avoids placing the service times of different areas on top of each other.

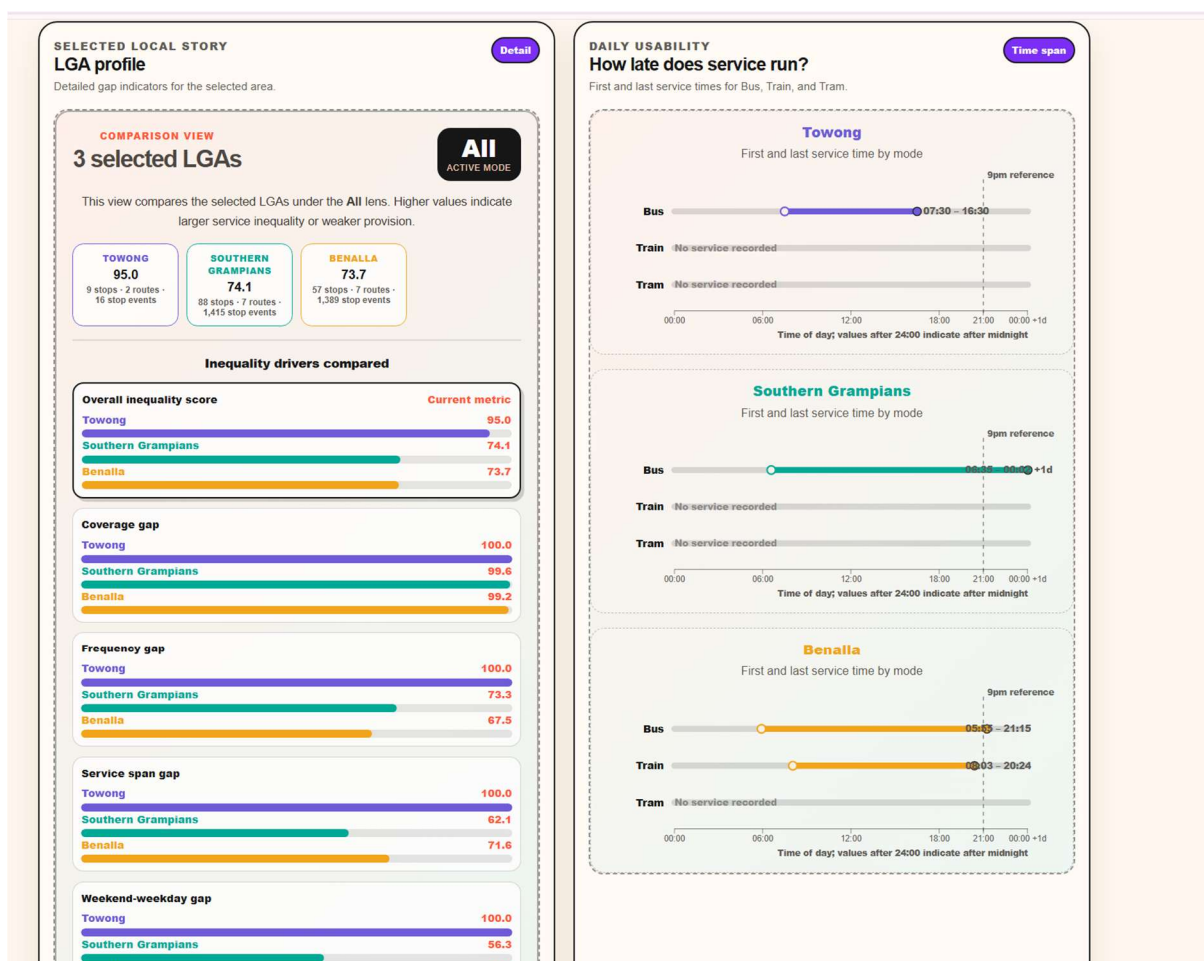


Figure 3 LGA Profile (Left) & Service Span Chart (Right)

The third group of views (see Figure 4 below) is used to show deeper service patterns. The weekday versus weekend dumbbell chart compares stop events on weekdays and weekends. This design uses two end points and a connecting line to show the gap between weekday and weekend service levels. It helps users see that some areas may still have a certain level of service on weekdays, but their

weekend service drops clearly. The chart shows the current Top N areas. It also adds selected LGAs that are not included in the Top N. Because of this, a selected area will not disappear from the analysis view just because it has a lower ranking. This design keeps the narrative continuous. Once users select an area, all later views continue to explain that choice.

The bubble chart summarises the position of different LGAs from the view of coverage and service intensity. The x-axis uses stop density, and the y-axis uses events per stop. The bubble size shows GTFS stop events. The colour shows the currently selected inequality metric. This design changes the earlier sketch idea of using population size. Instead, it uses stop events, which are more closely related to public transport service supply. Through this chart, users can identify different situations. For example, some areas may have high stop density but low service intensity. Other areas may have many service events but limited coverage. This helps users understand the difference between access and usability in more detail.

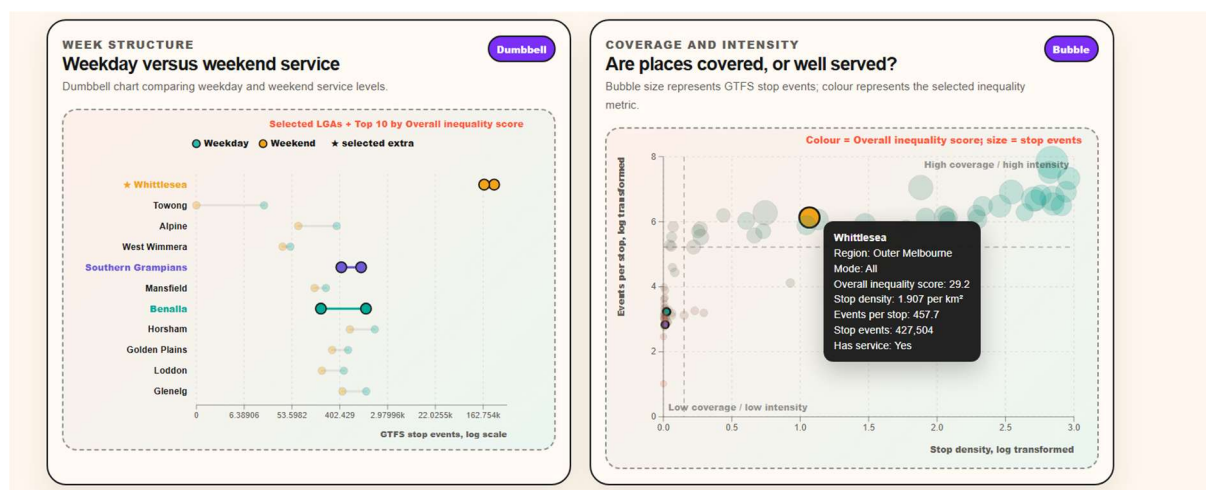


Figure 4 Dumbbell Chart (Left) & Bubble Chart (Right)

Finally, the conclusion matrix is used to compress the previous indicators into a summary explanation (see Figure 5 below). Based on the service features of the selected LGA, it places the area into different summary positions. This helps users judge whether inequality mainly comes from limited coverage, weak frequency, short service span, poor weekend availability, or lack of mode diversity. This matrix is not designed to replace the earlier charts. Instead, it works as the end of the narrative. It helps users form an overall judgement from several linked views. It also responds to the question raised at the beginning. An area may look like it has public transport. However, what really affects

daily life is whether the service is frequent, whether it runs late enough, whether weekend service is reliable, and whether different modes together form a usable network.



Figure 5 Conclusion matrix

Using the Implementation

The interactive narrative visualisation of this project needs to be run through a local server. It should not be opened by directly double-clicking the index.html file. To run the project, users should first unzip the submitted code folder. They should check that it contains index.html, css/style.css, js/main.js, data/processed/lga_metrics.csv, and data/processed/vic_lga_2025_simplified.geojson. Then, users can start a local server inside the project folder. For example, they can use the Live Server extension in VS Code.

One important interaction that can be easily missed is that this visualisation supports the selection of up to three LGAs for comparison. Users can click different areas on the map or in the ranking chart. The selected areas are highlighted with different colours. The same colours are kept across all charts. This helps users track the same LGA across different views. If users select more than three areas, the earliest selected area will be replaced automatically. Clicking an already selected LGA again will remove it from the selection.

Another function that can be easily missed is that the map supports zoom and pan. Users can use the mouse wheel to zoom in or out. They can also drag the map to view local areas. This is especially useful for dense areas such as Inner Melbourne. The zoom and pan functions are implemented with D3's built-in zoom behaviour. No extra map library is needed. Users can also use the Reset view button to return to the default settings. This includes the region, mode, metric, Top N selection, and the currently selected LGAs.

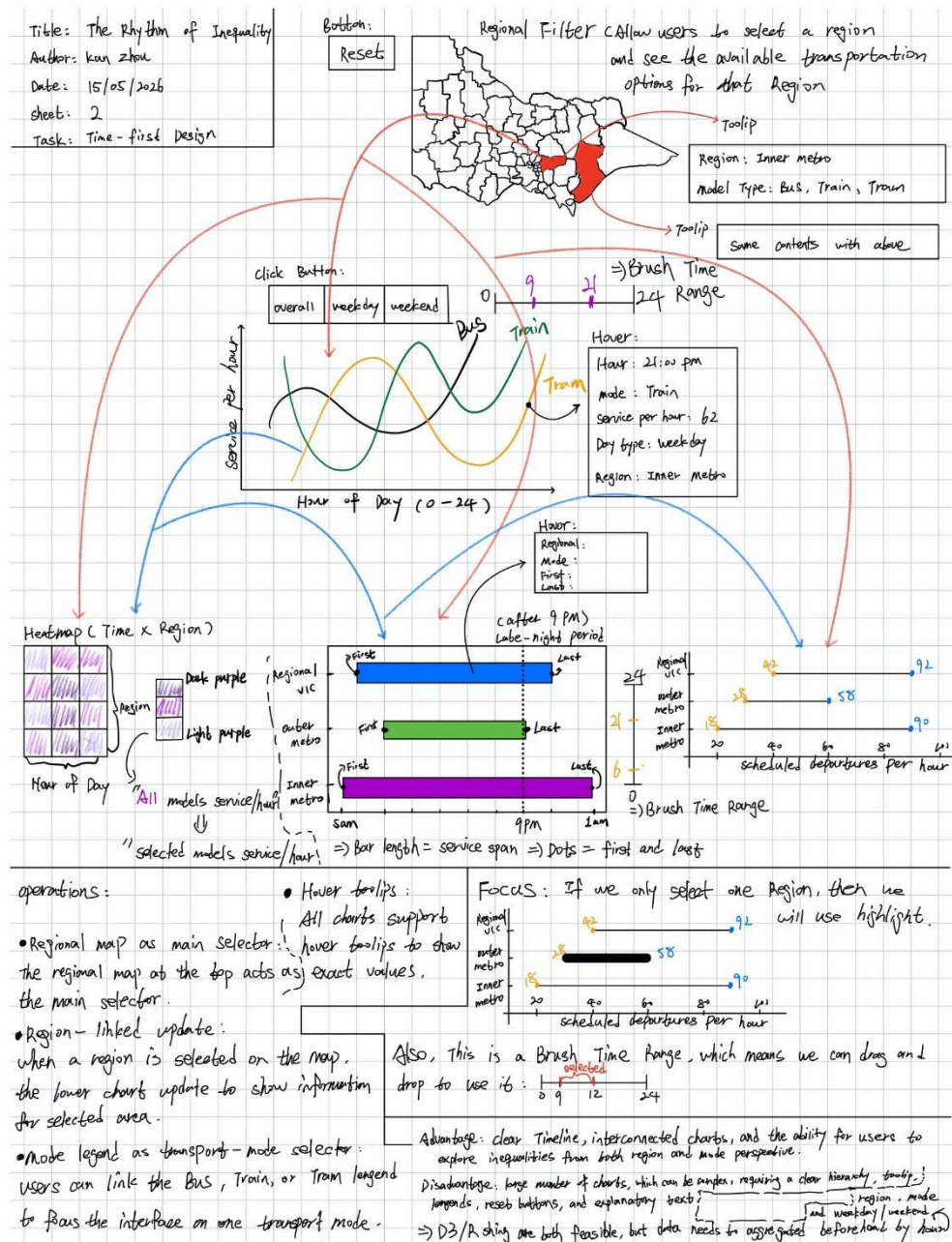
Conclusion and Reflection

This project aims to show that public transport inequality is not only about whether an area has stations, stops, or routes. More importantly, it is about whether these services are truly usable in daily life. Through the final interactive narrative visualisation, users can see that public transport service gaps between different LGAs in Victoria are multi-layered. Some areas may look covered by the public transport network in the data. However, the real user experience may still be limited by low service frequency, short operating hours, weak weekend service, or limited transport mode choices. The map and ranking chart show the spatial distribution of service gaps. They also show which LGAs are the most underserved. The LGA profile, service span chart, weekday-weekend comparison, bubble chart, and conclusion matrix then further explain the reasons behind these gaps.

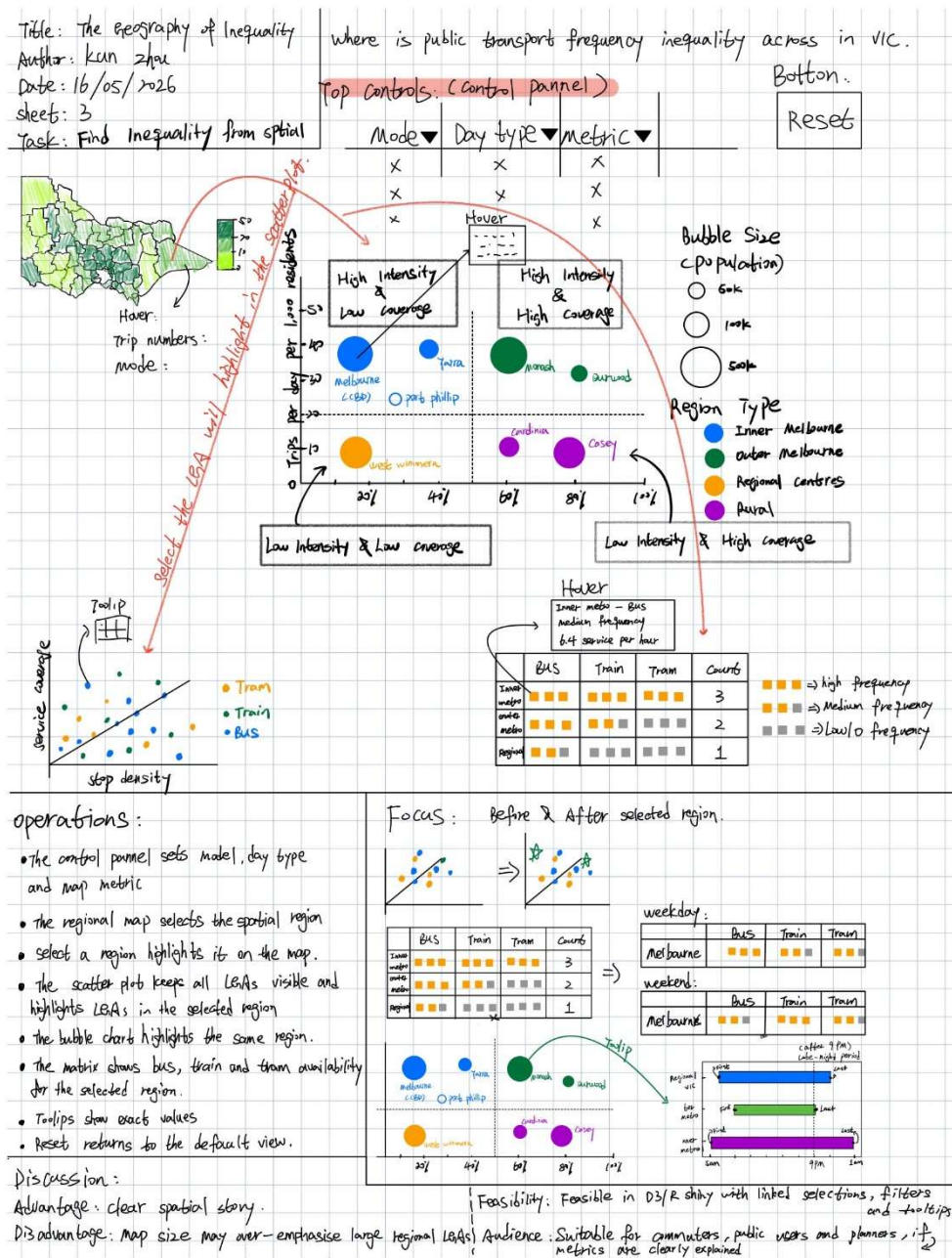
One important lesson I learned from this project is that narrative visualisation does not only need correct charts. It also needs data, interaction, and text explanations to work together. I also realised that many small details in interaction design can directly affect whether users understand the story smoothly. For example, after users zoom in on the map, selecting an area should not automatically return the map to the original view. The selection colours for multiple LGAs should stay consistent across all charts. Comparison views should avoid too much overlap. The areas selected by users should also remain visible in later charts.

Looking back at the whole process, if I had more time, I would continue to improve the project in several ways. First, I would test the layout more systematically on different screen sizes. This is especially important when showing the Top 20 LGAs and selecting multiple areas at the same time, because some charts may become crowded. Second, I would further expand the data model. I would add demand-side indicators, such as population, employment density, or social service needs. This would allow the visualisation to show not only public transport supply, but also the relationship between supply and potential demand. Third, I would add clearer annotations or a step-by-step guide. This would help first-time users find some less obvious interactions, such as selecting up to three LGAs, clicking again to remove a selection, and using zoom and pan on the map.

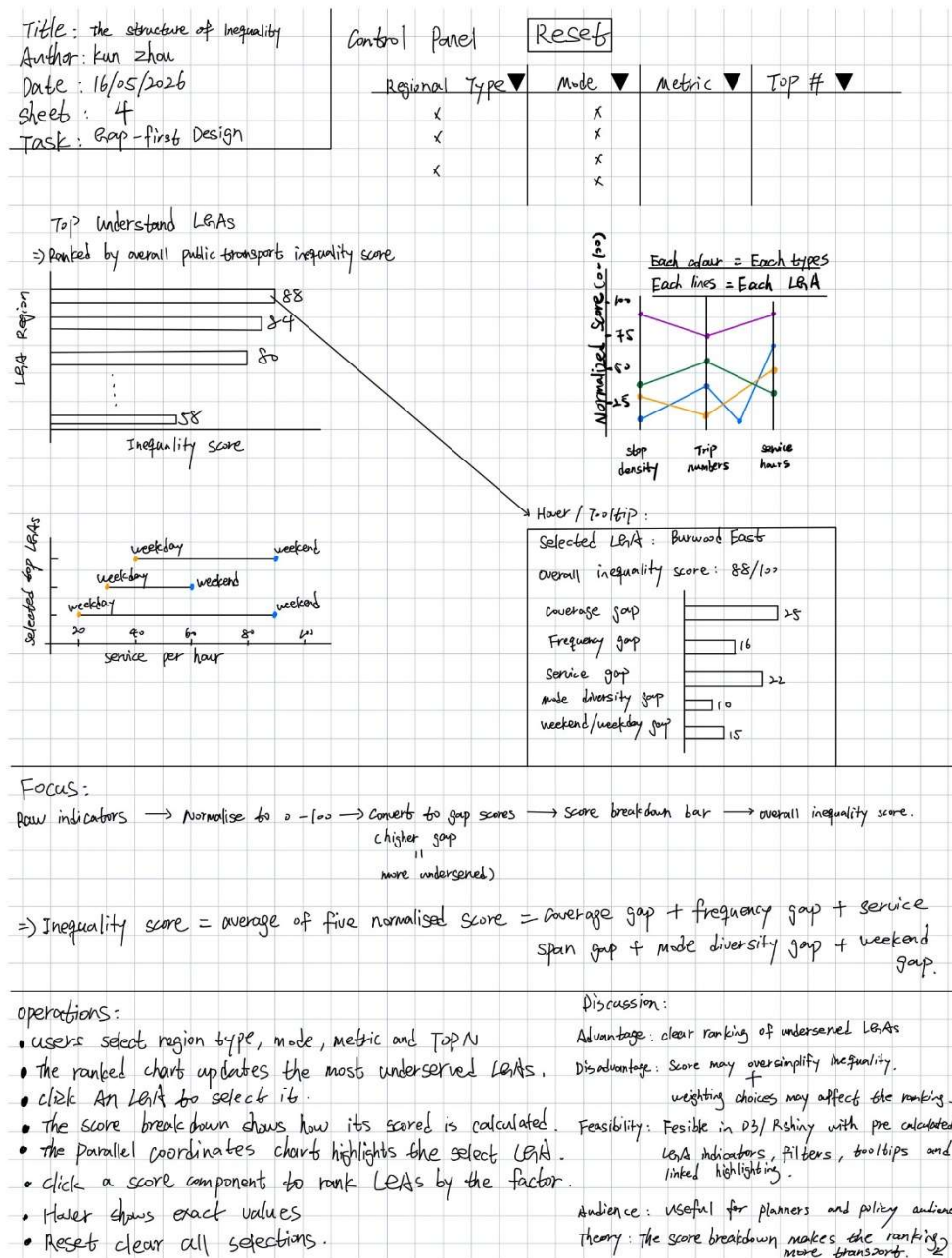
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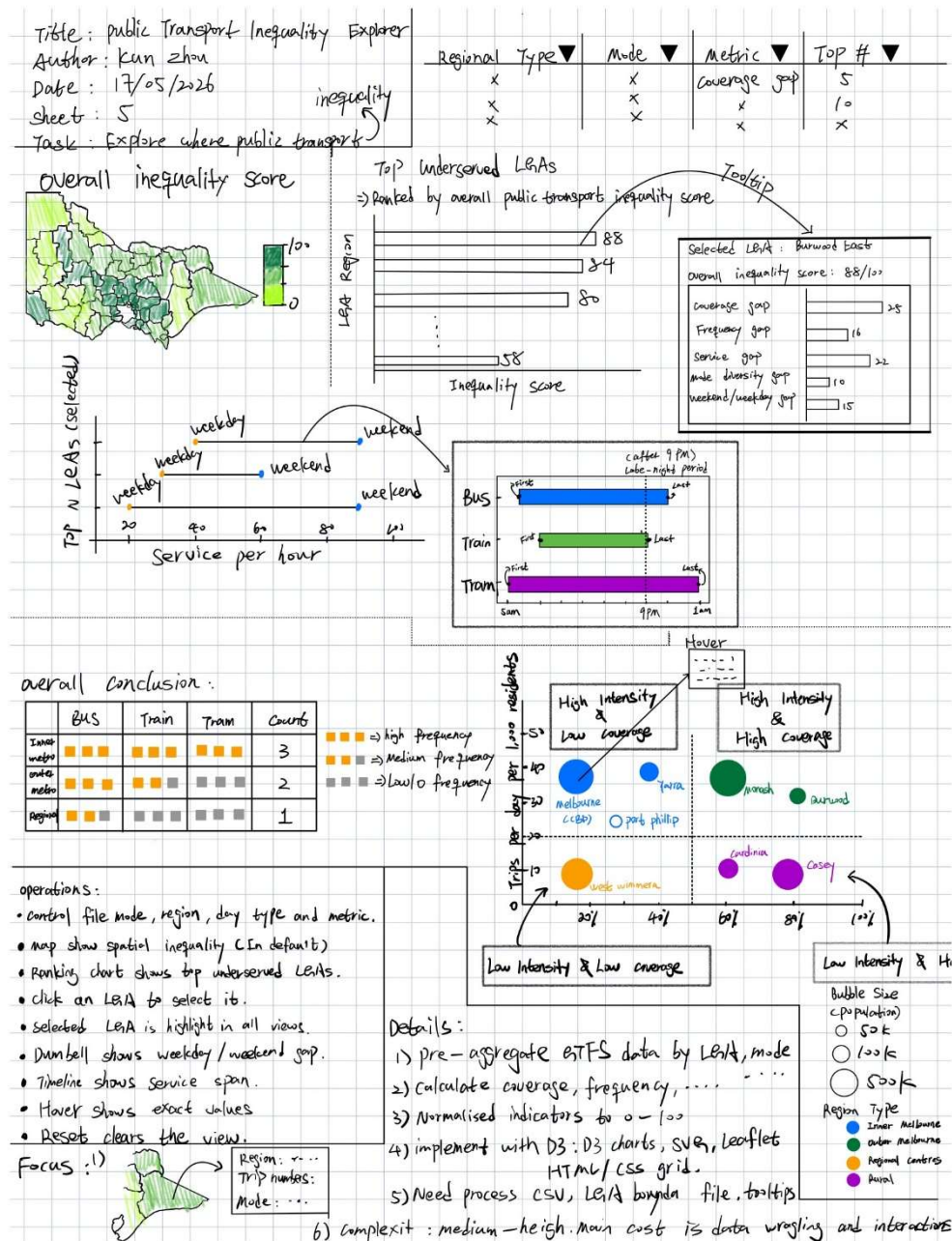
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